



Investment in the mining industry: Sustainable education and green literacy concepts

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ABSTRACT

This paper examines the impact of the Sustainable Education Index on China's mining industry, focusing on investment patterns and the adoption of green technologies. Using data from 1992 to 2022 and estimating with the Generalized Method of Moments (GMM), our findings reveal that a higher Sustainable Education Index significantly boosts green technology imports while reducing investment in mining activities. Specifically, a 1% increase in the index is associated with a 0.23% decrease in investment and a 0.43% increase in green technology imports. Additionally, the COVID-19 pandemic negatively impacted both investment and technology imports. While GDP and population growth positively correlate with investment, they are negatively associated with green technology imports, highlighting the challenges of urban development. An increased carbon footprint drives green technology imports, whereas a higher consumer price index deters both investment and imports. To promote sustainable investment, we recommend practical policies such as social awareness campaigns, education initiatives, university incentives, and social media engagement to guide China's mining industry toward greater environmental responsibility and sustainability.

1. Introduction

Investing in sustainability within the mining industry is essential for promoting a responsible and eco-friendly approach to resource extraction. According to Tu et al. (2020), Yoshino et al. (2021), and Famiyeh et al. (2021), sustainable mining practices aim to minimize environmental impact while maximizing social and economic benefits. This involves adopting innovative technologies and complying with strict environmental regulations to lower the carbon footprint of mining operations (Cardano and Zevallos, 2021; Budiono and Purba, 2023; Cheng et al., 2024). Since the metals and mining industry accounts for 4%–7% of global greenhouse gas emissions, it is essential to invest in initiatives that support sustainable practices.

Social awareness is crucial for encouraging the adoption of sustainable practices in the mining industry. As consumers become more informed about the environmental and social impacts of their purchases, demand for sustainably sourced commodities is rising (Piselli et al., 2022). By educating the public on sustainable mining practices and the importance of ethical sourcing, individuals and organizations can influence market trends and shape consumer behavior (Taghizadeh-Hesary et al., 2022; Verrier et al., 2022; Zhao and

Rasoulinezhad, 2023; Pouresmaeli et al., 2023). This increased awareness drives companies to invest in sustainable mining initiatives, ensuring they meet consumer expectations and maintain their social license to operate (Rasoulinezhad and Taghizadeh-Hesary, 2022; Agrawal et al., 2023).

Green literacy and sustainable education are essential for raising social awareness about sustainability issues, including those related to the mining industry. By equipping individuals with the knowledge and skills to understand the interconnections between environmental, social, and economic systems, green literacy promotes informed decision-making and responsible citizenship. Sustainable education empowers people to recognize the environmental impacts of their actions and encourages the adoption of sustainable lifestyles and consumption patterns. In the mining industry context, green literacy helps individuals understand the environmental challenges of resource extraction and the importance of sustainable mining practices. Additionally, sustainable education fosters environmental stewardship, motivating individuals to support policies and initiatives that prioritize environmental protection and social equity.

China faces significant challenges in achieving sustainability as the world's largest emitter of CO₂ and the top consumer of fossil fuels. Its

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vast and dominant mining industry leads in producing various minerals, contributing substantially to the country's GDP. However, the scale of mining operations exacerbates environmental degradation and carbon emissions (Guo and Yang, 2023; Xue et al., 2024). The mining sector alone significantly contributes to China's CO₂ emissions, with a marked increase between 2000 and 2014. Despite efforts to transition to cleaner energy sources and enforce stricter environmental regulations, the magnitude of China's mining industry presents a severe challenge to sustainability efforts (Lin et al., 2023). Addressing these challenges requires concerted efforts from both government and industry stakeholders to prioritize sustainable practices, invest in cleaner technologies, and reduce reliance on fossil fuels (Ye and Rasoulinezhad, 2023; Li et al., 2023a,b).

This paper investigates how the Sustainable Education Index impacts two critical aspects of the mining industry: investment trends and the adoption of green technologies. The index includes indicators like state investment in education, enrollment of rural students in higher education, and the number of educational institutions, reflecting a country's commitment to sustainable education practices. By analyzing the relationship between this index and investment patterns in mining, the study aims to understand how education policies influence investment decisions in the industry. Additionally, it explores how the Sustainable Education Index correlates with importing green technologies for mining equipment such as automatic trucks and drilling systems. Based on these aims, the hypothesis of the paper is: "A higher Sustainable Education Index positively influences investment trends in the mining industry and the adoption of green technologies. Specifically, countries with greater state investment in education, higher enrollment of rural students in higher education, and more educational institutions will likely increase investment in mining and import more green technologies for mining equipment."

This research adopts novel approaches to examine the unique dynamics of the Chinese mining industry and assess the influence of sustainable education on its trajectory. By focusing specifically on China, this study acknowledges the country's pivotal role as both a major player in global mining and a significant emitter of CO₂ emissions. Through an interdisciplinary lens that integrates insights from education policy, environmental science, and economic analysis, the research aims to unravel the complex interplay between sustainable education initiatives and mining practices in China. By leveraging comprehensive datasets and employing advanced analytical methods, this study seeks to uncover previously overlooked correlations and causal relationships between education investments, mining industry behavior, and the adoption of green technologies.

The research is structured as follows: Section 1 introduces the research objectives. Section 2 provides a brief literature review, identifying gaps in existing knowledge. Section 3 offers background information and theoretical discussions. Data sources and methodology are detailed in Section 4. Section 5 presents the findings, analyzing the impacts of sustainable education on the Chinese mining industry. Finally, Section 6 concludes the paper, summarizing key findings and suggesting directions for future research.

2. Literature review

In today's world, there is a resounding consensus among scholars, policymakers, and environmentalists that sustainability stands as the sole viable path toward safeguarding our environment and finite resources. In this context, the following literature serves as a representative sample.

The intersection of the mining industry and sustainable development has received considerable scholarly attention in recent years, highlighting growing awareness of the sector's significant environmental, social, and economic impacts. Studies such as those by Asr et al. (2019), Phung et al. (2023) and Chen et al. (2023) have focused on the environmental consequences of mining, including habitat destruction, water

contamination, and deforestation, emphasizing the urgent need for sustainable mining practices. Meanwhile, research by Rasoulinezhad (2020), and Frederiksen and Banks (2022) has examined the social aspects of mining, including indigenous rights, community well-being, and labor conditions, advocating for inclusive decision-making processes and equitable benefit-sharing mechanisms. Additionally, economic analyses by Monteiro et al. (2019) have assessed mining's role in economic development, addressing challenges such as resource dependency and the promotion of alternative livelihoods.

Sustainable education and green literacy have emerged as pivotal components in addressing the pressing environmental challenges of our time. This burgeoning field encompasses a diverse array of educational approaches, curricula, and initiatives aimed at fostering an understanding of ecological principles, environmental stewardship, and sustainable lifestyles. Drawing upon insights from scholars such as Dong et al. (2023), Chen and Ding (2023), and Hu and Zheng (2023), sustainable education seeks to cultivate a deep appreciation for the interconnectedness of human societies and the natural world, empowering learners to become active agents of positive change in their communities and beyond. Through experiential learning, interdisciplinary coursework, and engagement with real-world issues, students develop critical thinking skills, environmental awareness, and a sense of responsibility towards future generations. Moreover, green literacy initiatives, as outlined by authors like Nousheen et al. (2020), and Abo-Khalil (2024), aim to equip individuals with the knowledge, skills, and values needed to make informed decisions that promote sustainability in everyday life.

The sustainability of the mining industry in China represents a multifaceted challenge amid the country's rapid economic development and resource demands. Research by authors such as Yu et al. (2008), and Li et al. (2023) underscores the environmental degradation caused by extensive mining activities, including soil and water pollution, habitat destruction, and air quality deterioration. These impacts are exacerbated by the prevalence of outdated technologies and lax enforcement of environmental regulations. Moreover, the social dimensions of mining, including land rights disputes, labor rights violations, and community displacement, pose significant challenges to sustainability, as highlighted in studies by Jiao et al. (2021), and Yang et al. (2024). However, amid these challenges, as confirmed by earlier studies such as Chen et al. (2022), and Li et al. (2023), there are also promising signs of progress. Government initiatives such as the Circular Economy Development Strategy and the Green Mining Action Plan demonstrate a growing recognition of the need for sustainable mining practices. Additionally, collaborative efforts between industry stakeholders, academia, and civil society organizations are driving innovations in environmental management, community engagement, and corporate responsibility within the sector.

In conclusion, while the literature on the sustainability of the mining industry in China provides valuable insights into the environmental, social, and economic challenges facing the sector, there remains a noticeable gap in understanding the influence of sustainable education on critical aspects such as investment trends and the adoption of green technologies. Existing research highlights the urgency of addressing environmental degradation, social conflicts, and economic dependencies associated with mining activities. However, there is limited empirical evidence exploring the role of sustainable education in shaping industry practices and decision-making processes. Therefore, this paper aims to investigate the influence of the Sustainable Education Index on two critical aspects of the mining industry: investment trends and the adoption of green technologies. The research hypothesis posits that a higher level of sustainable education among industry stakeholders will positively correlate with increased investment in environmentally responsible mining projects and the adoption of green technologies, ultimately contributing to the long-term sustainability and resilience of the mining sector in China.

3. Theoretical background

China's mining industry commands a formidable presence globally, characterized by extensive mineral resources and substantial production capabilities. With 173 minerals discovered and proven reserves in 163 minerals, China ranks third worldwide, trailing only the United States and Russia in mineral endowment. Notably, China holds significant dominance in critical minerals, boasting more than 50% of global processing capacity for lithium, nickel, cobalt, and graphite. Forecasts for the Mining & Quarrying market project robust growth, with production anticipated to reach 5007.0 billion kilograms by 2024, supported by an expected annual growth rate of 5.64% (CAGR, 2024–2028). China's mining sector is intricately linked with national policy objectives, including the shift towards green, low-carbon industries outlined in China's Policy Strategies for Green Low Carbon goals proposed in 2021. Aligned with these objectives, China's National Plan for Mineral Resources (2016–2020) emphasizes the imperative of green transformation in resource development, advocating for sustainable practices to ensure the long-term environmental and economic sustainability of the mining industry.

Enhancements in sustainable education have the potential to catalyze a transformative shift towards green literacy and heightened sustainability awareness within the mining industry. A robust theoretical framework is crucial to understand the channels through which sustainable education influences mining practices and environmental outcomes. The transmission channels of impacts are as follows:

Enhanced Understanding of Environmental Impacts: Through sustainable education, individuals develop a nuanced understanding of the environmental impacts of mining, such as habitat destruction, water pollution, and carbon emissions. This awareness enables them to identify and support practices that mitigate these impacts, fostering a culture of environmental responsibility within the industry.

Recognition of the Value of Sustainable Practices: Sustainable education helps stakeholders appreciate the long-term advantages of adopting sustainable practices, including reduced environmental impact, improved community well-being, and enhanced regulatory compliance. By recognizing these benefits, stakeholders are more likely to advocate for and invest in sustainable mining projects.

Informed Decision-Making: A deeper understanding of sustainability principles empowers investors and industry leaders to make informed decisions that prioritize environmental responsibility. This shift in decision-making is driven by the recognition of the intrinsic link between responsible mining practices and long-term socio-environmental prosperity.

Culture of Innovation and Accountability: Sustainable education fosters a culture of innovation and accountability by equipping stakeholders with the knowledge to drive positive change. This culture is characterized by a commitment to reducing pollution, fostering inclusive development, and adhering to stringent environmental regulations.

Based on these transmission channels, we can select GDP, population, carbon footprint, consumer price index, and the impact of COVID-19 as key indicators for evaluating the effectiveness of sustainable education in the mining industry. GDP serves as an indicator of economic activity and growth. It reflects the overall economic performance of a country and can provide insights into how sustainable practices influence economic outcomes in the mining sector. The population size and growth can impact and be impacted by mining activities. Sustainable education may affect how mining operations address population-related challenges, such as resource demand and environmental impacts. The carbon footprint measures the total greenhouse gas emissions resulting from mining activities. Sustainable education aims to reduce these emissions by promoting environmentally responsible practices. Monitoring changes in the carbon footprint provides a direct assessment of how well education initiatives are reducing the environmental impact of mining. The CPI tracks changes in the price level of a basket of goods and services over time. It is a measure of inflation and can indicate how

sustainable practices in mining influence the cost of resources and products. The COVID-19 pandemic has had significant effects on global industries, including mining. Assessing the impact of COVID-19 helps understand how sustainable education might influence the resilience of mining operations in the face of global disruptions. This indicator can shed light on how well-prepared and adaptable the industry is to crises, reflecting the effectiveness of sustainability initiatives.

4. Research material

In this paper, the variables are carefully defined and selected to investigate the influence of the Sustainable Education Index on two crucial aspects of the mining industry: investment trends and the adoption of green technologies. The dependent variables in the empirical models are identified as investment in the mining industry and imports of green technologies within the mining sector. The primary explanatory variable under scrutiny is the Sustainable Education Index. The study employs a time series analysis focusing on China over a span of 30 years, from 1992 to 2022. This timeframe is chosen due to data availability and the significant policy shift catalyzed by China's Ten Strategic Policies on Environment and Development published in 1992, marking a pivotal moment in China's commitment to sustainable development. Additionally, a dummy variable representing the impact of COVID-19 is incorporated, taking a value of 1 for the years 2020, 2021, and 2022, and 0 for other years, acknowledging the pandemic's disruptive effects on the Chinese economy. The inclusion of this variable aims to provide valuable insights into the pandemic's repercussions on mining industry investment and the adoption of green technologies in China. Furthermore, several general variables, including GDP, population, carbon footprint, and consumer price index, are selected as control series in the empirical model to account for broader economic and environmental factors. A comprehensive summary of the description and data for all selected variables is presented in Table 1 for reference and clarity.

The expected impact of the Sustainable Education Index, COVID-19 impact, GDP, Population, Carbon Footprint, and Consumer Price Index on Investment in the mining industry and Imports of green technologies in the mining industry can be outlined as follows:

A higher SEI indicates a stronger focus on sustainable practices and education within the society. A higher SEI is likely to correlate positively with both investment in the mining industry and the adoption of green

Table 1
Description of variables.

Variable	Abbreviation	Unit	Observation	Sources
Sustainable education index	SEI	Index	1992–2022	Authors using data from the China Statistical Yearbook
Investment in mining industry	INV-MI	Million CNY	1992–2022	China Statistical Yearbook
Imports of green technologies in mining industry	GT-MI	Million CNY	1992–2022	Trademap
COVID-19 impact	COVID19	Dummy [0,1]	1992–2022	WHO
GDP	GDP	Million CNY	1992–2022	The China Statistical Yearbook
Population	POP	Number	1992–2022	The China Statistical Yearbook
Carbon footprint	CF	Metric tons CO2	1992–2022	BP, IEA
Consumer price index	CPIN	Index	1992–2022	World Bank

Source: Authors

technologies. This is because companies and investors are increasingly considering sustainability factors in their decision-making processes, which may lead to higher investments in sustainable mining practices and technologies. In addition, the COVID-19 pandemic has had profound effects on global economies, including the mining industry. The pandemic disrupted supply chains, slowed economic activity, and led to fluctuations in commodity prices. Consequently, the impact of COVID-19 on investment in the mining industry and imports of green technologies is expected to be negative during the years of peak disruption (2020–2022), as indicated by the dummy variable. However, the long-term effects may vary depending on recovery efforts and changes in market dynamics. GDP serves as a measure of the overall economic performance of a country. A growing GDP suggests increased economic activity, which can drive demand for raw materials and minerals, consequently influencing investment in the mining industry. Moreover, a higher GDP may signify greater financial resources available for investing in green technologies within the mining sector, aligning with sustainable development goals. Moreover, population growth can drive demand for resources, infrastructure, and energy, all of which are closely linked to the mining industry. A larger population typically implies higher consumption levels, leading to increased demand for products derived from mining activities. Additionally, population growth may necessitate investments in sustainable practices to mitigate environmental impacts and ensure resource sustainability, thereby affecting both investment in the mining industry and the adoption of green technologies. The carbon footprint reflects the environmental impact of economic activities, including those related to mining and industrial processes. Governments and stakeholders worldwide are increasingly prioritizing environmental sustainability and carbon reduction efforts. A higher carbon footprint may lead to regulatory pressures and public scrutiny, prompting investments in cleaner technologies and sustainable practices within the mining industry. Additionally, changes in the CPI reflect inflationary pressures and shifts in purchasing power. Inflation can affect production costs, profitability, and investment decisions within the mining sector. Additionally, fluctuations in the CPI may influence consumer behavior and demand for eco-friendly products, thereby indirectly impacting the adoption of green technologies in mining.

The estimation process commences with the construction of a correlation matrix to assess the relationships and potential multicollinearity among the variables. This step provides initial insights into the interdependencies within the dataset. Following this, two stationary tests, namely the Augmented Dickey Fuller and Phillips-Perron tests, are employed to determine the integration orders of the time series data. These tests help identify whether the variables exhibit unit roots, indicating non-stationarity, or are stationary in nature. Subsequently, the cointegration analysis proposed by Johansen is conducted to ascertain whether the variables are cointegrated in the long term, implying a stable relationship among them. This step is crucial for understanding the existence of equilibrium relationships among the variables. Finally, the estimation of coefficients is carried out using the generalized method of moments (GMM). This method is preferred due to its robustness to heteroscedasticity and distributional assumptions, which are common in economic datasets. The equations representing the relationships between the variables are formulated based on GMM principles, with specific equations dedicated to the two dependent variables under investigation, as denoted by Equations (1) and (2), respectively.

$$INV_MI_t = \beta_0 + \beta_1 INV_MI_{t-1} + \beta_2 SEI_t + \beta_3 GDP_t + \beta_4 POP_t + \beta_5 CF_t + \beta_6 CPIN_t + \beta_7 COVID_{dummy} + error\ term_t \quad [1]$$

$$GT_MI_t = \beta_0 + \beta_1 GT_MI_{t-1} + \beta_2 SEI_t + \beta_3 GDP_t + \beta_4 POP_t + \beta_5 CF_t + \beta_6 CPIN_t + \beta_7 COVID_{dummy} + error\ term_t \quad [2]$$

5. Findings

The empirical analysis findings are presented as follows. Table 2 displays the correlation matrix using a pairwise comparison approach. With p-values significant at 5% and values less than 0.5, it is evident that collinearity is absent within our empirical model (see Table 3).

Subsequently, the integration orders for all series are assessed by conducting unit root tests, namely ADF and PP tests as utilized in this study. The results are presented in Table 4, indicating that the variables exhibit varying patterns of stationarity.

Following this, an examination is conducted to determine whether the variables become cointegrated in the long term. For this purpose, the Johansen test is employed, and the results are presented in Table 4. Based on the analysis, it can be concluded that there is indeed a cointegration relationship among the variables.

The findings from the empirical analysis (Table 5) provide compelling evidence that the Sustainable Education Index serves as a significant motivator for imports of green technologies within the mining industry in China, while exerting a negative impact on investment in mining activities. The results indicate that a 1% improvement in the Sustainable Education Index correlates with a noteworthy 0.23% decrease in investment in the mining industry, which typically encompasses environmentally unfriendly practices. Conversely, the same improvement in the Sustainable Education Index corresponds to a substantial 0.43% increase in imports of green technologies, thus facilitating the greening of the mining sector in China. These outcomes can be interpreted through several primary reasons. Firstly, heightened awareness and education regarding sustainability issues incentivize stakeholders in the mining industry to prioritize the adoption of green technologies as a means of mitigating environmental impacts and complying with regulatory standards. Secondly, the negative impact on investment in mining may stem from increased scrutiny and pressure from regulatory bodies and investors, prompting a shift away from conventional mining practices towards more sustainable alternatives.

Furthermore, the findings affirm that the COVID-19 outbreak in China has led to a deceleration in both investment and imports of green technologies within the mining industry. This observation can be elucidated by the disruptive nature of the pandemic, which instigated widespread economic turmoil and uncertainty. The outbreak necessitated stringent containment measures, including lockdowns and restrictions on movement, which disrupted supply chains, hampered economic activities, and constrained investment opportunities. Consequently, the mining industry, like many other sectors, experienced a slowdown in investment as companies grappled with financial constraints and risk aversion in the face of uncertain market conditions. Similarly, the importation of green technologies was impeded as logistical challenges and disruptions in global trade hindered the

Table 2
Pairwise correlation matrix.

	SEI	INV_MI	GT_MI	GDP	POP	CF	CPIN
SEI	1						
INV_MI	−0.01 (0.00)	1					
GT_MI	0.16 (0.00)	0.08 (0.00)	1				
GDP	−0.06 (0.01)	0.31 (0.07)	0.05 (0.02)	1			
POP	−0.13 (0.03)	0.26 (0.06)	−0.15 (0.01)	−0.04 (0.00)	1		
CF	0.24 (0.00)	−0.22 (0.00)	0.38 (0.08)	0.16 (0.03)	0.09 (0.01)	1	
CPIN	−0.12 (0.09)	0.15 (0.07)	−0.26 (0.00)	0.26 (0.00)	−0.03 (0.00)	−0.30 (0.00)	1

Note: Dummy variable (COVID-19) can not be included to the correlation analysis; () are p-values.

Source: Authors

Table 3
Stationarity evaluation.

	ADF test		PP test	
	At level	At 1st difference	At level	At 1st difference
SEI	−2.59 (0.85)	−8.40 (0.00)	−1.51 (0.86)	−5.44 (0.00)
INV-MI	−6.10 (0.01)	−19.93 (0.00)	−4.39 (0.03)	−13.29 (0.00)
GT-MI	−0.034 (1.000)	−2.54 (0.92)	−0.055 (0.93)	−1.69 (0.96)
GDP	−1.16 (0.79)	−8.10 (0.00)	−1.28 (0.99)	−5.40 (0.00)
POP	−2.64 (0.88)	−10.59 (0.01)	−1.95 (0.73)	−6.18 (0.00)
CF	−4.38 (0.03)	−16.50 (0.00)	−7.10 (0.02)	−18.52 (0.00)
CPIN	−0.072 (0.79)	−1.30 (0.64)	−0.030 (0.95)	−1.28 (0.73)

Note: Dummy variable (COVID-19) can not be included to the stationary analysis; () are p-values.

Source: Authors

Table 4
Cointegration evaluation outcome.

Vector		Eigen-value	Max-Eigen statistic	Critical value at 5%
H0	H1			
$r = 0$	$r > 0$	0.643	45.603	42.640
$r \leq 1$	$r > 1$	0.492	21.338	24.396
$r \leq 2$	$r > 2$	0.410	11.550	16.171
$r \leq 3$	$r > 3$	0.083	0.516	2.695

Source: Authors

Table 5
GMM results.

Variables	Dependent variable: Investment in mining industry		Dependent variable: Imports of green technologies in mining industry	
	Coefficient	P-value	Coefficient	P-value
Investment in mining industry (−1)	0.154	0.026	–	–
Imports of green technologies in mining industry (−1)	–	–	0.241	0.479
Sustainable education index	−0.231	0.009	0.436	0.019
COVID-19 impact	−0.081	0.031	−0.092	0.069
GDP	0.493	0.031	−0.297	0.012
Population	0.103	0.004	−0.112	0.063
Carbon footprint	0.021	0.194	0.319	0.040
Consumer price index	−0.159	0.001	−0.148	0.031
Observations	30		30	
Number of instruments	12		14	

Source: Authors

procurement and transportation of environmentally friendly equipment and technologies. Additionally, the focus of governments and industries shifted towards immediate crisis management and recovery efforts, diverting attention and resources away from sustainability initiatives.

Regarding GDP and population, the estimated coefficients confirm that a 1% increase in these variables correlates with an increase in investment in the mining industry and a decrease in imports of green technologies within the sector. These findings shed light on the prevailing lack of an adequate green growth concept and insufficient green literacy, particularly in the context of urban development and sustainability initiatives in China. The observed positive relationship between GDP and investment in the mining industry underscores the traditional economic growth model prioritizing resource extraction and industrial expansion. As GDP increases, there is a corresponding rise in demand for raw materials and infrastructure development, driving investment in mining activities. However, this growth model often disregards

environmental concerns and sustainability considerations, leading to heightened exploitation of natural resources and environmental degradation. Simultaneously, the negative relationship between GDP (and population) and imports of green technologies highlights a systemic deficiency in prioritizing environmentally sustainable practices. Despite economic growth and population expansion, the adoption of green technologies within the mining industry remains limited. This suggests a gap in understanding and embracing sustainable development principles, as well as inadequate investment in eco-friendly infrastructure and technology adoption.

The findings reveal that a 1% increase in carbon footprint solely exerts a positive impact on imports of green technologies within the mining industry in China, while exhibiting no significant effect on investment in mining activities. This outcome can be elucidated by the increasing global emphasis on environmental sustainability and the growing recognition of the imperative to mitigate carbon emissions. A higher carbon footprint indicates greater environmental degradation and pollution, prompting regulatory bodies, investors, and industry stakeholders to prioritize green technologies and eco-friendly solutions as a means of mitigating environmental impact. The positive impact of carbon footprint on imports of green technologies suggests a reactive response to environmental concerns, wherein industries seek to adopt cleaner technologies to address environmental degradation. As the carbon footprint increases, there is a corresponding rise in demand for green technologies to reduce environmental impact and comply with regulatory standards. This aligns with broader sustainability trends and reflects efforts to transition towards greener and more environmentally responsible practices within the mining industry. However, the lack of a significant impact on investment in mining industry despite the increase in carbon footprint may indicate a disconnect between environmental considerations and investment decisions. It suggests that while environmental concerns may influence technology adoption, they may not necessarily drive overall investment behavior in the mining sector.

Lastly, the findings indicate that the Consumer Price Index (CPI) exerts negative impacts on both investment in the mining industry and imports of green technologies within the sector in China. This observation can be interpreted through several primary reasons. Firstly, a negative relationship between CPI and investment in the mining industry may reflect the influence of inflationary pressures on production costs and profitability. As CPI increases, companies in the mining sector face higher operating expenses, reducing their capacity or willingness to invest in new projects or expansion initiatives. Additionally, rising consumer prices may dampen demand for mined commodities, further deterring investment in the sector. Similarly, the negative impact of CPI on imports of green technologies suggests that inflationary pressures may impede the adoption of environmentally friendly technologies within the mining industry. Higher consumer prices may constrain companies' budgets for capital expenditures, limiting their ability to invest in costly green technologies. Moreover, elevated CPI levels may signal broader economic challenges, such as reduced purchasing power and weakened consumer confidence, which could translate into decreased demand for green technologies. Furthermore, the negative relationship between CPI and green technologies imports underscores the importance of affordability and cost-effectiveness in driving technology adoption. Inflationary pressures may exacerbate the financial barriers associated with acquiring and implementing green technologies, particularly for smaller firms or those operating on tight budgets. As a result, companies may prioritize short-term cost considerations over long-term sustainability goals, delaying or scaling back investments in green technologies.

If we consider the dependent variable of imports of green technologies in the mining industry as a proxy for greening the industry, and the dependent variable of investment in the mining industry as a proxy for browning the industry, the comparisons of impacts of explanatory variables reveal that sustainable education is an efficient way to promote greening the mining industry in China while curbing the expansion of

browning the industry in the country. This observation can be elucidated through several key factors. Firstly, sustainable education plays a pivotal role in raising awareness and fostering a culture of environmental stewardship within the mining sector. By equipping industry stakeholders with the knowledge and skills needed to understand and implement sustainable practices, sustainable education empowers them to prioritize the adoption of green technologies and eco-friendly solutions. This promotes the greening of the mining industry by encouraging investments in cleaner technologies, resource efficiency, and environmental management practices. Conversely, the negative impact of sustainable education on investment in the mining industry suggests that heightened awareness of sustainability issues may deter or redirect investment away from traditional, environmentally unfriendly practices. As stakeholders become more cognizant of the environmental and social impacts associated with mining activities, they may exhibit greater reluctance to invest in projects that perpetuate environmental degradation or social harm. This serves to mitigate the expansion of browning the industry, wherein investment is primarily directed towards conventional mining practices with little regard for sustainability considerations.

Table 6 displays the results of diagnostic tests designed to verify the suitability of the empirical models. The values shown confirm that the models do not exhibit problems such as serial correlation or over-identification. This indicates that the models effectively capture the relationships between variables without significant biases or distortions from these common statistical issues. Specifically, the absence of serial correlation suggests that the error terms are not systematically correlated over time, while the lack of overidentification means that the models are neither overparameterized nor affected by multicollinearity.

6. Conclusion and policy recommendations

In this paper, we meticulously define and select variables to explore how the Sustainable Education Index influences two crucial aspects of China's mining industry: investment trends and the adoption of green technologies. Our empirical models designate investment in the mining sector and imports of green technologies as dependent variables, with the Sustainable Education Index serving as the primary explanatory variable. Spanning a 30-year time series from 1992 to 2022, our analysis provides compelling evidence that improvements in the Sustainable Education Index significantly drive up imports of green technologies while concurrently reducing investment in mining activities. Specifically, a 1% enhancement in the Sustainable Education Index correlates with a notable 0.23% decline in mining industry investment and a substantial 0.43% increase in green technology imports. These findings underscore the pivotal role of sustainable education in incentivizing the adoption of green technologies and steering the mining industry towards environmentally responsible practices.

Furthermore, our study confirms the adverse impacts of the COVID-19 pandemic on both investment and green technology imports within China's mining sector. The disruptions caused by the pandemic, including supply chain interruptions and economic uncertainty, have contributed to a slowdown in investment and technology adoption, reflecting broader challenges faced by the industry in crisis management and recovery efforts.

Additional analysis reveals that GDP and population growth positively correlate with investment in the mining sector but negatively impact green technology imports, highlighting urban development and industry practices that often overlook sustainability considerations. Conversely, increases in carbon footprint positively influence green technology imports, signaling a responsive approach to environmental concerns. However, rises in the consumer price index negatively affect both investment and green technology imports, illustrating affordability challenges and cost considerations that hinder sustainability initiatives.

Finally, by using imports of green technologies as an indicator of industry greening and investment as a measure of non-sustainable

Table 6

Diagnostic tests.

Test	Dependent variable: Investment in mining industry		Dependent variable: Imports of green technologies in mining industry	
	Value	P-value	value	P-value
AR (1)- serial correlation	−2.154	0.032	−2.662	0.028
AR (2)- serial correlation	−0.327	0.794	−0.417	0.196
Over identification test (Sargan)	19.031	0.612	17.923	0.498
Wald test	893.493	0.000	962.102	0.001

Source: Authors

practices, our comparisons underscore the effectiveness of sustainable education in promoting greening while mitigating non-sustainable activities within China's mining sector. Sustainable education empowers stakeholders to prioritize eco-friendly practices, thereby redirecting investment towards sustainability and away from environmentally harmful activities.

To enhance sustainable investment in China's mining industry and align with the broader goals of sustainability, several practical policies can be implemented. Firstly, nationwide campaigns should be launched to raise social awareness of the environmental and social impacts of mining, encouraging stakeholders to prioritize sustainability in their decisions. Developing education programs focused on sustainable development, particularly in less developed regions, can address knowledge gaps and empower local communities to adopt eco-friendly practices. Incentivizing academic and industry collaboration, especially through programs aligned with international agreements like the Rio +20 Declaration, can drive innovation in green technologies and sustainable practices. Additionally, enhancing green literacy through digital platforms and social media can reach a broader audience, promoting the importance of sustainability across various sectors. Finally, introducing fiscal and regulatory incentives for sustainable investment, such as tax breaks or subsidies for green technologies, can attract more investment into sustainable mining projects, ensuring they are both economically viable and environmentally responsible.

In the future, to build upon the contributions of this paper, researchers could explore the requirements for a sustainable mining industry in China in the post-COVID era. This research would involve examining the evolving dynamics of the mining sector following the pandemic, identifying emerging challenges, and outlining strategies for incorporating sustainability principles into industry practices. Additionally, investigating financing solutions for sustainable investments in the mining industry represents another promising area for further study.

CRedit authorship contribution statement

Jie Li: Writing – review & editing, Writing – original draft, Resources, Methodology, Data curation, Conceptualization. **Xu Zou:** Data curation, Conceptualization.

Declaration of competing interest

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all the authors. We declare that there is no conflict of interest.

Data availability

Data will be made available on request.

References

- Abo-Khalil, A., 2024. Integrating sustainability into higher education challenges and opportunities for universities worldwide. *Heliyon* 10 (9), e29946. <https://doi.org/10.1016/j.heliyon.2024.e29946>.
- Agrawal, M., Kalia, P., Nema, P., Zia, A., Kaur, K., John, H., 2023. Evaluating the influence of government initiatives and social platforms on green practices of Gen Z: the mediating role of environmental awareness and consciousness. *Cleaner and Responsible Consumption* 8, 100109. <https://doi.org/10.1016/j.clrc.2023.100109>.
- Asr, E., Kakaie, R., Ataei, M., Mohammadi, M., 2019. A review of studies on sustainable development in mining life cycle. *J. Clean. Prod.* 229, 213–231.
- Burdiono, S., Purba, J., 2023. Factors of foreign direct investment flows to Indonesia in the era of COVID-19 pandemic. *Heliyon* 9 (4), e15429. <https://doi.org/10.1016/j.heliyon.2023.e15429>.
- Chen, J., Wen, S., Liu, Y., 2022. Research on the efficiency of the mining industry in China from the perspective of time and space. *Resour. Pol.* 75, 102475 <https://doi.org/10.1016/j.resourpol.2021.102475>.
- Chen, X., Ding, H., 2023. Assessment of green economic development through the impacts of fossil fuels resources and education in China. *Resour. Pol.* 85 (Part B), 103910 <https://doi.org/10.1016/j.resourpol.2023.103910>.
- Chen, Y., He, M., Xu, Y., 2023. Sustainable development of the mining sector for achieving common prosperity in Chinese rural areas. *Resour. Pol.* 87 (Part A), 104325 <https://doi.org/10.1016/j.resourpol.2023.104325>.
- Cheng, W., Dilanchiev, A., Mammadov, E., Wanjun, S., 2024. The impact of natural resource extensive on green economic growth policies. *Resour. Pol.* 92, 104980 <https://doi.org/10.1016/j.resourpol.2024.104980>.
- Cordano, A., Zevallas, R., 2021. Country competitiveness and investment allocation in the mining industry: a survey of the literature and new empirical evidence. *Resour. Pol.* 73, 102136 <https://doi.org/10.1016/j.resourpol.2021.102136>.
- Dong, X., Zhang, X., Zhang, C., Bi, C., 2023. Building sustainability education for green recovery in the energy resource sector: a cross country analysis. *Resour. Pol.* 81, 103385 <https://doi.org/10.1016/j.resourpol.2023.103385>.
- Famiyeh, S., Opoku, R., Kwarteng, A., Asante-Darko, D., 2021. Driving forces of sustainability in the mining industry: evidence from a developing country. *Resour. Pol.* 70, 101910 <https://doi.org/10.1016/j.resourpol.2020.101910>.
- Frederiksen, T., Banks, G., 2022. How would a mining sector that was committed to sustainable development behave: a 21st century manifesto. *Extr. Ind. Soc.* 11, 101116 <https://doi.org/10.1016/j.exis.2022.101116>.
- Guo, Y., Yang, F., 2023. Mining safety research in China: understanding safety research trends and future demands for sustainable mining industry. *Resour. Pol.* 83, 103632 <https://doi.org/10.1016/j.resourpol.2023.103632>.
- Hu, N., Zheng, B., 2023. Natural resources, education, and green economic development. *Resour. Pol.* 86 (Part B), 104053 <https://doi.org/10.1016/j.resourpol.2023.104053>.
- Jiao, W., Zhang, X., Li, C., Guo, J., 2021. Sustainable transition of mining cities in China: literature review and policy analysis. *Resour. Pol.* 74, 101867 <https://doi.org/10.1016/j.resourpol.2020.101867>.
- Li, Y., Pinto, M., Kumar, D., 2023a. Analyzing sustainability indicator for Chinese mining sector. *Resour. Pol.* 80, 103275 <https://doi.org/10.1016/j.resourpol.2022.103275>.
- Li, Y., Zhang, Y., Yang, L., Du, F., Sai, L., Zhang, B., 2023b. Decoupling analysis of China's mining industrial development and water usage: based on production-based and consumption-based perspectives. *J. Clean. Prod.* 385, 135668 <https://doi.org/10.1016/j.jclepro.2022.135668>.
- Lin, W., Wang, P., Yuan, M., 2023. Governing the knowledge commons: hybrid relational-contractual governance in China's mining industry. *World Dev.* 172, 106376 <https://doi.org/10.1016/j.worlddev.2023.106376>.
- Monteiro, N., Silva, E., Neto, J., 2019. Sustainable development goals in mining. *J. Clean. Prod.* 228, 509–520.
- Nousheen, A., Zai, S., Waseem, M., Khan, S., 2020. Education for sustainable development (ESD): effects of sustainability education on pre-service teachers' attitude towards sustainable development (SD). *J. Clean. Prod.* 250, 119537 <https://doi.org/10.1016/j.jclepro.2019.119537>.
- Phung, T., Rasoulinezhad, E., Thu, H., 2023. How are FDI and green recovery related in Southeast Asian economies? *Econ. Change Restruct.* 56, 3735–3755.
- Piselli, C., Colladon, A., Segneri, L., Pisello, A., 2022. Evaluating and improving social awareness of energy communities through semantic network analysis of online news. *Renew. Sustain. Energy Rev.* 167, 112792 <https://doi.org/10.1016/j.rser.2022.112792>.
- Pouresmaeili, M., Ataei, M., Qarahasanlou, A., barabadi, A., 2023. Integration of renewable energy and sustainable development with strategic planning in the mining industry. *Results in Engineering* 20, 101412. <https://doi.org/10.1016/j.rineng.2023.101412>.
- Rasoulinezhad, E., 2020. Environmental impact assessment analysis in the kahak's wind farm. *J. Environ. Assess. Pol. Manag.* 22 (1), 2250006 <https://doi.org/10.1142/S1464333222500065>.
- Rasoulinezhad, E., Taghizadeh-Hesary, F., 2022. Role of green finance in improving energy efficiency and renewable energy development. *Energy Efficiency* 15 (14). <https://doi.org/10.1007/s12053-022-10021-4>.
- Taghizadeh-Hesary, F., Phoumin, H., Rasoulinezhad, E., 2022. COVID-19 and regional solutions for mitigating the risk of SME finance in selected ASEAN member states. *Econ. Anal. Pol.* 74, 506–525.
- Tu, C., Rasoulinezhad, E., Sarker, T., 2020. Investigating solutions for the development of a green bond market: evidence from analytic hierarchy process. *Finance Res. Lett.* 34, 101457 <https://doi.org/10.1016/j.frl.2020.101457>.
- Verrier, B., Smith, C., Yahyaie, M., Ziemiński, M., Forbes, G., Witt, K., Azadi, M., 2022. Beyond the social license to operate: whole system approaches for a socially responsible mining industry. *Energy Res. Social Sci.* 83, 102343 <https://doi.org/10.1016/j.erss.2021.102343>.
- Xue, L., Li, M., Zheng, Z., Xi, S., Yang, Y., Yang, W., Hou, Q., 2024. Assessing the progress of the mining industry towards green growth in China: a three-stage dynamic network slacks-based measure approach. *J. Clean. Prod.* 435, 140478 <https://doi.org/10.1016/j.jclepro.2023.140478>.
- Yang, S., Jahangir, A., Balsalobre-Lorente, D., 2024. Sustainable resource management in China's energy mining sector: a synthesis of development and conservation in the FinTech era. *Resour. Pol.* 89, 104665 <https://doi.org/10.1016/j.resourpol.2024.104665>.
- Ye, X., Rasoulinezhad, E., 2023. Assessment of impacts of green bonds on renewable energy utilization efficiency. *Renew. Energy* 202, 626–633.
- Yoshino, N., Rasoulinezhad, E., Taghizadeh-Hesary, F., 2021. Economic impacts of carbon tax in a general equilibrium framework: empirical study of Japan. *J. Environ. Assess. Pol. Manag.* 23 (1), 2250014 <https://doi.org/10.1142/S1464333222500144>.
- Yu, J., Zhang, Z., Zhou, Y., 2008. The sustainability of China's major mining cities. *Resour. Pol.* 33 (1), 12–22.
- Zhao, L., Rasoulinezhad, E., 2023. Role of natural resources utilization efficiency in achieving green economic recovery: evidence from BRICS countries. *Resour. Pol.* 80, 103164 <https://doi.org/10.1016/j.resourpol.2022.103164>.